LAB: Principal Component Analysis

2026-02-27

```r
# We will use the following packages.
# If needed, install them : pak::pkg_install().
stopifnot(
  require("corrr"),
  require("magrittr"),
  require("lobstr"),
  require("ggforce"),
  require("gt"),
  require("glue"),
  require("skimr"),
  require("patchwork"),
  require("tidyverse"),
  require("ggfortify")
  # require("autoplotly")
)
```

```r
old_theme <- theme_set(theme_minimal())

options(ggplot2.discrete.colour="viridis")
options(ggplot2.discrete.fill="viridis")
options(ggplot2.continuous.fill="viridis")
options(ggplot2.continuous.colour="viridis")
```

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Swiss fertility data

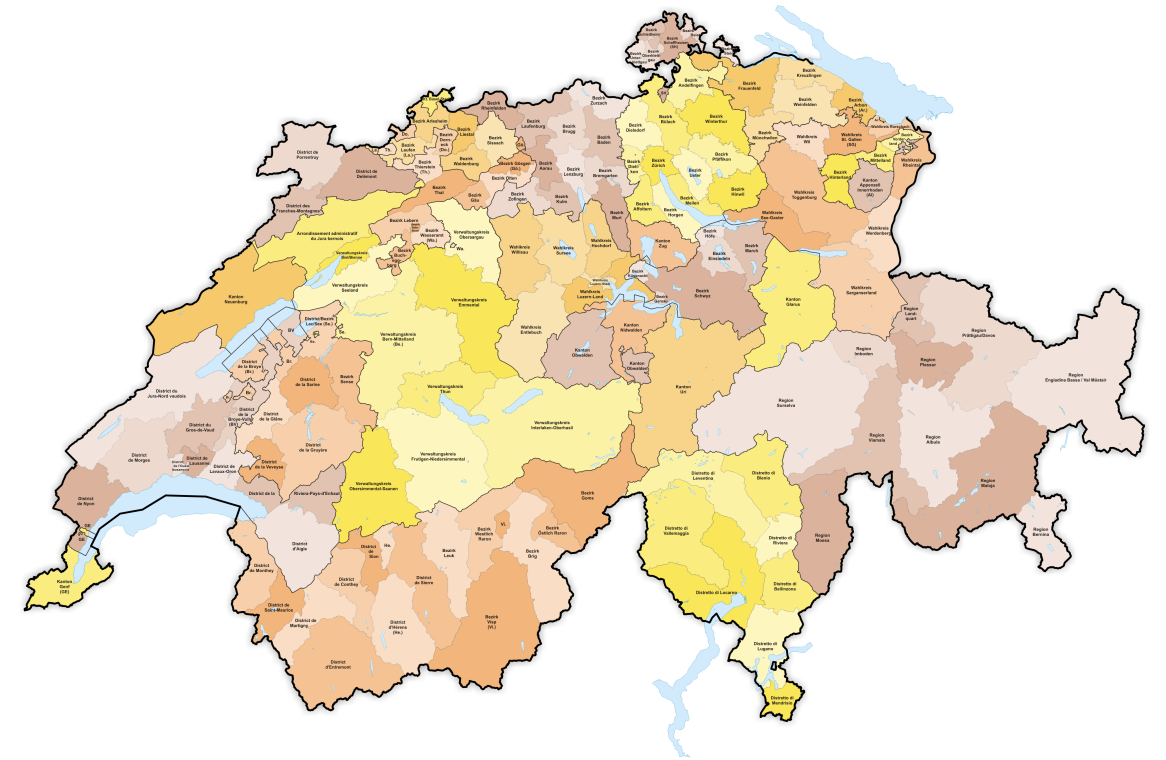
Dataset `swiss` from `datasets::swiss` connect fertility and social, economic data within 47 French-speaking districts in Switzerland.

- `Fertility` : fertility index
- `Agriculture` : jobs in agricultural sector
- `Examination` : literacy index (military examination)
- `Education` : proportion of people with successful secondary education
- `Catholic` : proportion of Catholics
- `Infant.Mortality` : mortality quotient at age 0

Fertility index (`Fertility`) is considered as the *response variable*

The social and economic variables are *covariates* (*explanatory* variables).

See European Fertility Project for more on this dataset.



PCA (Principal Component Analysis) is concerned with covariates.

```
data("swiss")

swiss %>%
  glimpse(50)
```

```
Rows: 47
Columns: 6
$ Fertility        <dbl> 80.2, 83.1, 92.5, 85.8,~
$ Agriculture      <dbl> 17.0, 45.1, 39.7, 36.5,~
$ Examination      <int> 15, 6, 5, 12, 17, 9, 16~
$ Education        <int> 12, 9, 5, 7, 15, 7, 7, ~
$ Catholic         <dbl> 9.96, 84.84, 93.40, 33.~
$ Infant.Mortality <dbl> 22.2, 22.2, 20.2, 20.3,~
```

Have a look at the documentation of the dataset

Describe the dataset

> **Question**
>
> Compute summary for each variable

solution

It is enough to call `summary()` on each column of `swiss`. This can be done in a functional programming style using package `purrr`. The collections of summaries can be rearranged so as to build a dataframe that is fit for reporting.

```
tt <- map_dfr(swiss, summary, .id = "var")
```

```
tt %>%
  knitr::kable()
```

var	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
Fertility	35.00	64.700	70.40	70.14255	78.450	92.5
Agriculture	1.20	35.900	54.10	50.65957	67.650	89.7
Examination	3.00	12.000	16.00	16.48936	22.000	37.0
Education	1.00	6.000	8.00	10.97872	12.000	53.0
Catholic	2.15	5.195	15.14	41.14383	93.125	100.0
Infant.Mortality	10.80	18.150	20.00	19.94255	21.700	26.6

Function `skim` from `skimr` delivers all univariate summaries in suitable form.

```
foo <- swiss %>%
  select(-Fertility) %>%
  skim()
```

```
foobar <- foo %>%
  filter(skim_type=="numeric") %>%
  rename(variable=skim_variable)  %>%
    mutate(across(where(is.numeric), ~ round(.x, digits=1)))
```

```
foobar %>%
  gt::gt()
```

skim_type	variable	n_missing	complete_rate	numeric.mean	numeric.sd	num
numeric	Agriculture	0	1	50.7	22.7	
numeric	Examination	0	1	16.5	8.0	
numeric	Education	0	1	11.0	9.6	
numeric	Catholic	0	1	41.1	41.7	
numeric	Infant.Mortality	0	1	19.9	2.9	

Question

Display graphic summary for each variable.

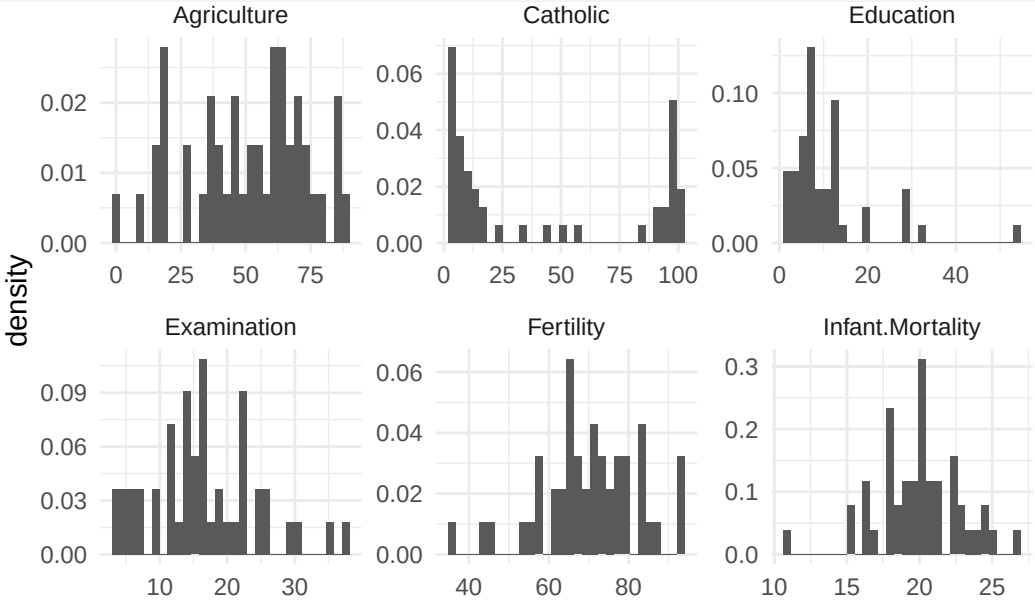
solution

We have to pick some graphical summary of the data. Boxplots and violine plots could be used if we look for concision.
We use histograms to get more details about each column.
Not that covariates have different meanings: `Agriculture`, `Catholic`, `Examination`, and `Education` are percentages with values between 0 and 100.
We have no details about the standardized fertility index `Fertility`
`Infant.Mortality` is also a rate:

> Infant mortality is the death of an infant before his or her first birthday. The infant mortality rate is the number of infant deaths for every 1,000 live births. In addition to giving us key information about maternal and infant health, the infant mortality rate is an important marker of the overall health of a society.

see Center for Desease Control
We reuse the function we have already developped during previous sessions.

```
make_biotifoul(swiss, .f = is.numeric)
```



Histograms reveal that our covariates have very different distributions.
Religious affiliation (`Catholic`) tells us that there two types of districts, which is reminiscent of the old principle *Cujus regio, ejus religio* , see Old Swiss Confederacy.
`Agriculture` shows that in most districts, agriculture was still a very important activity.
`Education` reveals that in all but a few districts, most children did not receive secondary education. `Examination` shows that some districts lag behind the bulk of districts. Even less exhibit a superior performance.
The two demographic variables `Fertility` and `Infant.Mortality` look roughly unimodal with a few extreme districts.

Investigate pairwise correlations

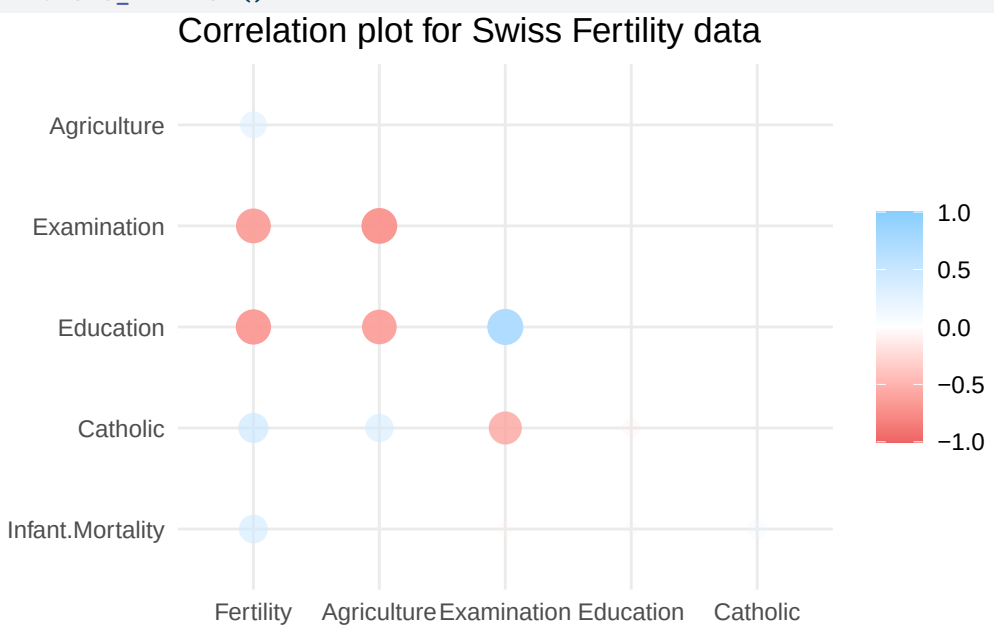
Question

- Compute, display and comment the sample correlation matrix
- Display jointplots for each pair of variables

solution

Package `corrr`, functions `correlate` and `rplot` provide a convenient tool.
Note that `corrr::rplot()` creates a graphical object of class `ggplot`. We can endow it with more layers.

```
swiss |>
    corrr::correlate(use="pairwise.complete.obs",method="pearson", quiet=T) |>
  corrr::shave() |>
  corrr::rplot() +
  labs(title="Correlation plot for Swiss Fertility data") +
  theme_minimal()
```



The high positive linear correlation between `Education` and `Examination` is moderately surprising. The negative correlation between the proportion of people involved in `Agriculture` and `Education` and `Examinationis` also not too surprising. Secondary schooling required pupils from rural areas to move to cities.
A more intriguing observation concerns the pairs `Catholic` and `Examination` (negative correlation) and `Catholic` and `Education` (little correlation).
The response variable `Fertility` looks negatively correlated with `Examination` an `Education`. These correlations are worth being further explored. In Demography, the decline of Fertility is often associated with the the rise of women education. Note that `Examination` is about males, and that `Education` does not give details about the way women complete primary education.

Singular Value Decomposition (SVD)

Question

- Project the `swiss` dataset on the covariates (all columns but `Fertility`)
- Center the projected data using matrix manipulation
- Center the projected data using `dplyr` verbs
- Compare the results with the output of `scale()` with various optional arguments
- Call the centered matrix `Y`

solution

Hand-made centering of the dataframe emphasises the fact that centering is a linear operation. As a matter of fact, it consists in projecting the data frame on the linear space orthogonal to the constant vector.

```
X <- select(swiss, -Fertility) |>
    as.matrix()

n <- nrow(X)
ones <-  matrix(1, nrow = n, ncol=1)

Y <-  X - (1/n)* (ones  %*%  t(ones) %*% X)
```

We can also perform centering using `dplyr` verbs. This can be viewed as computing a window function over a trivial partition.

```
swiss |>
  select(-Fertility) |>
  mutate(across(everything(), \(x) x-mean(x)))
```

Anyway, function `scale(X, scale=F)` from base `R` does the job.

Question

Check that the ouput of `svd(Y)` actually defines a Singular Value Decomposition.

solution

`svd(Y)` is a list with 3 elements (`u,d,v`).

$$Y = U \times D \times V^\top$$

```
svd_Y <-  svd(Y)

svd_Y %$%                                                    ①
  (Y - u %*% diag(d) %*% t(v)) %>%
  norm(type = "F")                                           ②

norm(
  diag(1, ncol(Y)) -
  (svd_Y %$% (t(v) %*% v)),
  'F' )                                                      ③
```

① Exposing pipe from `magrittr`
② Checking the factorization
③ Checking that colomns of `v` form an orthonormal family

```
[1] 1.04354e-13
[1] 1.137847e-15
```

Note that we used the exposing pipe `%$%` from `magrittr` to unpack `svd_Y` which is a list with class `svd` and members named `u`, `d` and `v`.
We could have used `with(,)` from base `R`.

Question

Relate the SVD of Y and the eigen decomposition of $Y^\top \times Y$

solution

The matrix $1/nY^\top \times Y$ is the covariance matrix of the covariates.
The spectral decomposition of the symmetric Semi Definite Positive (SDP) matrix $1/nY^\top \times Y$ is related with the SVD factorization of Y.
The spectral/eigen decomposition of $Y^\top \times Y$ can be obtained using `eigen()`.
The eigenspaces of $Y^\top \times Y$ are the right eigenspaces of Y.

```
(t(eigen(t(Y) %*% Y )$vectors) %*% svd_Y$v ) %>%
  round(digits=2)
```

```
     [,1] [,2] [,3] [,4] [,5]
[1,]    1    0    0    0    0
[2,]    0   -1    0    0    0
[3,]    0    0    1    0    0
[4,]    0    0    0    1    0
[5,]    0    0    0    0    1
```

The eigenvalues of $Y^\top \times Y$ are the squared singular values of Y

```
eigen(t(Y) %*% Y )$values - (svd_Y$d)^2
```

```
[1] -1.309672e-10 -1.455192e-11  8.640200e-12  8.526513e-12 -
3.410605e-13
```

Here, the eigenvectors of $Y^\top \times Y$ coincide with the right singular vectors of Y corresponding to non-zero singular values. Up to sign changes, it is always true when the non-zero singular values are pairwise distinct.

Perform PCA on covariates

Question

Pairwise analysis did not provide us with a clear and simple picture of the French-speaking districts.
PCA (Principal Component Analysis) aims at exploring the variations of multivariate datasets around their mean (center of inertia). In the sequel, we will perform PCA on the matrix of centered covariates, with and without standardizing the centered columns.
Base `R` offers `prcomp()`. Call `prcomp()` on the centered covariates
Note that `R` also offers `princomp`

We first call `prcomp()` with the default arguments for centering and scaling, that is, we center columns and do not attempt to standardize columns. Name the output `pco`.

What is the result made of?

solution

```r
pco <- swiss |>
  select(-Fertility) |>
  scale(scale = F) |>
  prcomp(scale. = F)
```

`pco` is a list with 5 members. It as a class attribute `prcomp`. It is an object of class `prcomp` (function `prcomp()` acts as a constructor for class `pco` just as `lm()` acts as a constructor for class `lm`). Class `pco` is an `S3` class

```r
rlang::is_list(pco)
```

```
[1] TRUE
```

```r
attributes(pco)
```

```
$names
[1] "sdev"     "rotation" "center"   "scale"    "x"

$class
[1] "prcomp"
```

```r
sloop::s3_class(pco)
```

```
[1] "prcomp"
```

Question

Check that `prcomp()` is indeed a wrapper for `svd()`.

solution

We first check that the matrix can be recovered from the product of the components of the `prcomp` object.

```
(Y - pco$x %*% t(pco$rotation )) |>
  round(digits = 2)  |>
  head()
```

```
             Agriculture Examination Education Catholic Infant.Mortality
Courtelary             0           0         0        0                0
Delemont               0           0         0        0                0
Franches-Mnt           0           0         0        0                0
Moutier                0           0         0        0                0
Neuveville             0           0         0        0                0
Porrentruy             0           0         0        0                0
```

We now check that the `rotation` component is indeed made of the right singular vectors (the V factor)

```
(svd_Y$v %*% t(pco$rotation )) %>%
  round(2)
```

```
     Agriculture Examination Education Catholic Infant.Mortality
[1,]           1           0         0        0                0
[2,]           0           1         0        0                0
[3,]           0           0         1        0                0
[4,]           0           0         0        1                0
[5,]           0           0         0        0                1
```

The column vectors of component x are pairwise orthogonal.

```
(t(pco$x) %*% pco$x) |>
  round(2)
```

```
         PC1      PC2     PC3    PC4    PC5
PC1 86484.49     0.00    0.00   0.00   0.00
PC2     0.00 21127.44    0.00   0.00   0.00
PC3     0.00     0.00 2706.14   0.00   0.00
PC4     0.00     0.00    0.00 639.22   0.00
PC5     0.00     0.00    0.00   0.00 348.01
```

The `x` component of the `prcomp` object is the product of the U and D factors.

```
norm(pco$x - svd_Y$u %*% diag(svd_Y$d),type="F")
```

```
[1] 1.962527e-13
```

```
norm(as.matrix(pco$rotation) -svd_Y$v, type="F")
```

```
[1] 2.04934e-15
```

The connection between `pco$sdev` and `svd_Y$d` is somewhat less transparent.

```
sum(abs(apply(pco$x, 2, sd) - pco$sdev))
```

```
[1] 1.865175e-14
```

The components of `pco$sdev` are the standard deviations of the columns of $U \times D$.

```
pco$sdev - svd_Y$d/sqrt(nrow(Y)-1)
```

```
[1] -7.105427e-15  0.000000e+00  8.881784e-16 -1.332268e-15 -
1.332268e-15
```

Question

Check that rows and columns of component `rotation` of the result of `prcomp()` have unit norm.

solution

Check that rows and columns of matrix `rotation` have unit norm.

```
apply(pco$rotation, 2, \(x) norm(x, "2"))
```

```
PC1 PC2 PC3 PC4 PC5
  1   1   1   1   1
```

```
apply(pco$rotation, 1, \(x) norm(x, "2"))
```

```
     Agriculture      Examination        Education         Catholic
               1                1                1                1
Infant.Mortality
               1
```

Question

Check Orthogonality of V (component `rotation` of the `prcomp` object)

solution

```r
# checking that pco$rotation is an orthogonal matrix
t(pco$rotation) %*% pco$rotation
```

```
              PC1           PC2           PC3           PC4           PC5
PC1  1.000000e+00 -1.003429e-16  8.239937e-18 -1.097213e-
16  6.938894e-18
PC2 -1.003429e-16  1.000000e+00  1.181780e-16  5.074066e-
17  4.857226e-17
PC3  8.239937e-18  1.181780e-16  1.000000e+00  1.717376e-16 -
6.938894e-17
PC4 -1.097213e-16  5.074066e-17  1.717376e-16  1.000000e+00 -
1.804112e-16
PC5  6.938894e-18  4.857226e-17 -6.938894e-17 -1.804112e-
16  1.000000e+00
```

```r
pco$rotation %*% t(pco$rotation)
```

```
                   Agriculture   Examination     Education      Catholic
Agriculture       1.000000e+00  3.642919e-17 -1.153591e-
16  1.689187e-16
Examination       3.642919e-17  1.000000e+00 -8.630249e-
17  2.244298e-17
Education        -1.153591e-16 -8.630249e-17  1.000000e+00 -
1.127570e-16
Catholic          1.689187e-16  2.244298e-17 -1.127570e-
16  1.000000e+00
Infant.Mortality  2.081668e-17 -1.734723e-16  8.326673e-17 -
2.081668e-17
                 Infant.Mortality
Agriculture          2.081668e-17
Examination         -1.734723e-16
Education            8.326673e-17
Catholic            -2.081668e-17
Infant.Mortality     1.000000e+00
```

Question

Make a scatterplot from the first two columns of the x component of the `prcomp` object.

solution

Objects of class `prcomp` can be handled by generic functions like `plot()` or better `autoplot()`. Namely, method `prcomp` for generic S3 function `autoplot()` from `ggplot2` delivers one of classical SVD plots.

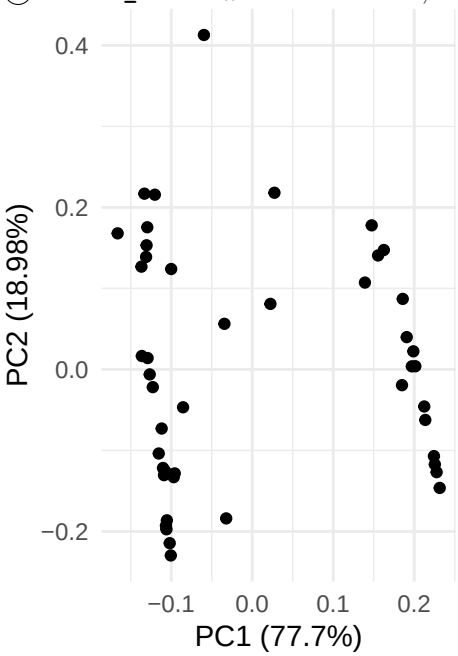
```
res <- autoplot(pco) +                                  ①
  coord_fixed() +                                       ②
  theme_minimal()

ts <- theme_set(theme_minimal())

res
```

① This call is dispatched to method `autoplot.prcomp` defined in package `ggfortify`. Look at the code using `getS3method("autoplot", "prcomp")`

② `coord_fixed()` is useful here, since scaling axes matter here.



`autoplot(pco)` is a scatterplot for the dataframe defined by matrix $U \times D$ projected on its first two principal components (first two columns).
As `autoplot(pco)` is an instance of class `ggplot`, it can be annotated, decorated as any other `ggplot` object.

```
(
  res + aes(color=Catholic) + theme_minimal()
) +
(
  res + aes(color=Education) + theme_minimal()
) +
  patchwork::plot_annotation(
    subtitle = "Scatterplot on the first two principal components (no column scaling)"
```

Question

Define a graphical pipeline for the *screeplot*.
Hint: use function `tidy()` from `broom`, to get the data in the right form from an instance of `prcomp`.

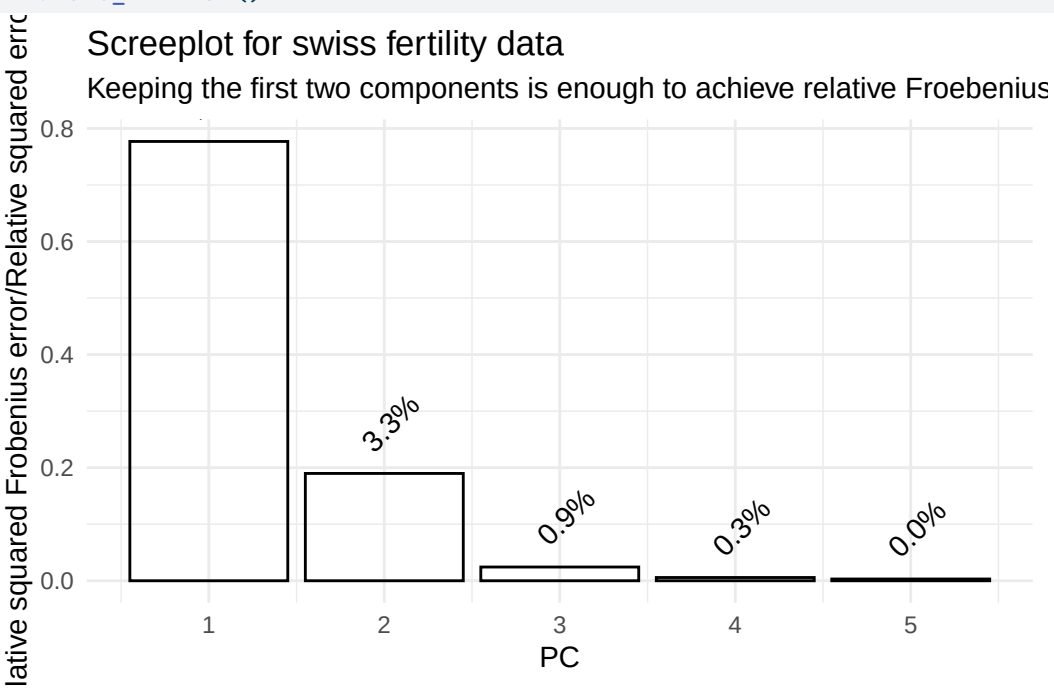
solution

The screeplot is a bar plot where each bar corresponds to a singular value. The bar height is proportional to the square of the corresponding singular value.

```
p_screeplot <- . %>%                                                    ①
  broom::tidy(matrix="pcs") %>%  {
  ggplot(.) +
  aes(x=PC, y=percent, label=pct_format(1.-cumulative)) +               ③
  geom_text(angle=45, vjust=-1, hjust=-.1) +
  geom_col(fill=NA, colour="black") +                                   ②
  theme_minimal()
  }
```

① Define a pipeline for building a screeplot (this works with `%>%` not with `|>`)
② Mind the braces on the right side of the first pipe
③ `1- cumulative` tell the reader about the relative Frobenious error achieved by keeping the first components of the SVD expansion.

```
pco %>%
  p_screeplot() +
  ylab('Relative squared Frobenius error/Relative squared error') +
  labs(
    title="Screeplot for swiss fertility data",
    subtitle="Keeping the first two components is enough to achieve relative Froebeniu
  theme_minimal()
```



The screeplot is a visualization of the Eckart-Young-Mirsky Theorem. It tells us about the relative errors incurent when approximating the data matrix (with centered columns) by the low rank approximations defined by the truncated SVDs.

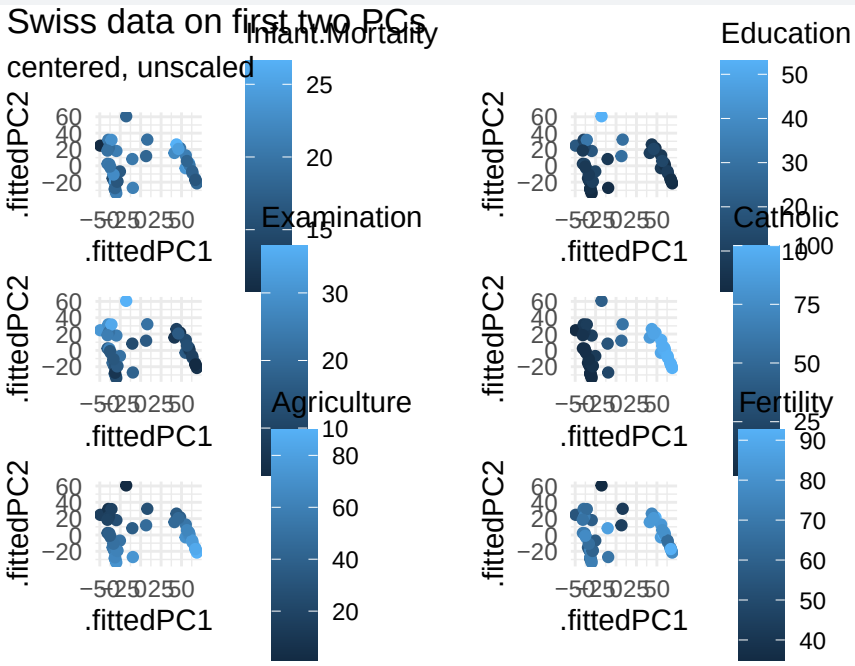
Question

Define a function that replicates `autoplot.prcomp()`
Project the dataset on the first two principal components (perform dimension reduction) and build a scatterplot. Colour the points according to the value of original covariates.
Hint: use generic function `augment()` from `broom`.

solution

```
p <-  pco %>%                                                    ①
  broom::augment(swiss) %>%
  ggplot() +
  aes(x=.fittedPC1, y=.fittedPC2, label=.rownames) +
  geom_point() +
  coord_fixed() +                                                ②
  ggrepel::geom_text_repel()  +
  theme_minimal()

(p +
  aes(color=Infant.Mortality)) +
(p +
   aes(color=Education)) +
(p +
   aes(color=Examination)) +
(p +
   aes(color=Catholic)) +
(p +
   aes(color=Agriculture)) +
(p +
   aes(color=Fertility)) +
plot_layout(ncol = 2) +
plot_annotation(title="Swiss data on first two PCs" ,
                subtitle = "centered, unscaled")
```



Question

Apply `broom::tidy()` with optional argument `matrix="v"` or `matrix="loadings"` to the `prcomp` object.
Comment.

solution

We can extract factor V from the SVD factorization using generic function `tidy` from package `broom`

```
pco %>%
  broom::tidy(matrix="v") %>%
  sample_n(5) |>
  gt::gt()  |>
  gt::fmt_number(decimals=2)
```

column	PC	value
Examination	4.00	-0.87
Education	3.00	-0.81
Catholic	1.00	0.95
Examination	2.00	0.17
Agriculture	3.00	-0.37

The result is a tibble in *long form*. It is worth pivoting the dataframe into wide form. This gives back the rotation matrix.

```
om <- pco %>%
  broom::tidy(matrix="v") %>%
  tidyr::pivot_wider(id_cols =column,
              names_from = PC,
              values_from = value) |>
  select(-1) |>
  as.matrix()

norm((om %*% t(om))-diag(1,5), "F")
```

```
[1] 8.196585e-16
```

Question

Build the third SVD plot, the so called *correlation circle*.

solution

The correlation circle is built from the *loadings*, that is, from the `rotation` component of the `prcomp` object.
We define a preprocessing function to transform the rotation object into a proper `tibble` form.

```
prep_co_circle <- function(pco) {
  r <- pco$rotation
  as_tibble(r) |>
    rename_with(.fn = \(x) gsub('PC', '', x), .cols=everything()) |>
    mutate(row_id=rownames(r))
}
```

The The next virtual graphical object will be our key tool to build the correlation circle.

```
co_circle_ppl <- (
    pco %>%
    prep_co_circle() %>%
    filter(F)
    ) %>%
  ggplot() +
  aes(x=`1`, y=`2`, label=row_id) +
  geom_segment(aes(xend=0, yend=0), arrow = grid::arrow(ends = "first")) +
  ggrepel::geom_text_repel() +
  coord_fixed() +                                                    (1)
  xlim(c(-1.1, 1.1)) + ylim(c(-1.1, 1.1))  +
  ggforce::geom_circle(aes(x0=0, y0=0, r=1), linetype="dashed") +
  theme_minimal()
```

(1) important

```
co_circle_ppl %+% (
  pco %>%
  prep_co_circle()
  )  +
  labs(title="Correlation circle",
          subtitle = "centered, unscaled",
          caption= "Swiss Fertility dataset") +
  theme_minimal()
```

```
Warning: <ggplot> %+% x was deprecated in ggplot2 4.0.0.
i Please use <ggplot> + x instead.
NULL
```

The length of each arrow is the length of the projection of the corresponding column of the data matrix over the plane generated by the first two rescaled left singular vectors (rescaling by the reciprocal of the singular values).
The first two principal componants (left singular vectors) are highly correlated with columns `Agriculture` and `Catholic`.

Question

Compute PCA after standardizing the columns, draw the correlation circle.

solution

```
pco2 <- select(swiss, -Fertility) |>
 prcomp(scale. = T)

co_circle_ppl %+% (
  pco2 %>%
  prep_co_circle()
  )  +
  labs(
    title="Correlation circle",
    subtitle = "centered, scaled",
    caption="Swiss fertility dataset") +
  theme_minimal()
```

```
NULL
```

Scaling columns seriously modify the correlation circle.

Compare standardized and non-standardized PCA

Question

Pay attention to the correlation circles.

1. How well are variables represented?
2. Which variables contribute to the first axis?

solution

```
pco_c <- swiss %>%
  select(-Fertility) %>%
  prcomp()

pco_cs <- swiss %>%
  select(-Fertility) %>%
  prcomp(scale.=T, center=T)
```

```
patchwork::wrap_elements(
(
  co_circle_ppl %+%
  prep_co_circle(pco_c)  +
  labs(
    subtitle = "centered, unscaled"
  ) +
  theme_minimal()
) ,
(
  co_circle_ppl %+%
  prep_co_circle(pco_cs) +
  labs(
    subtitle = "centered, scaled"
  ) +
  theme_minimal()
)) +
  patchwork::plot_annotation(
        title="Swiss, correlation circle"
  )
```

Swiss, correlation circle

Question

Explain the contrast between the two correlation circles.

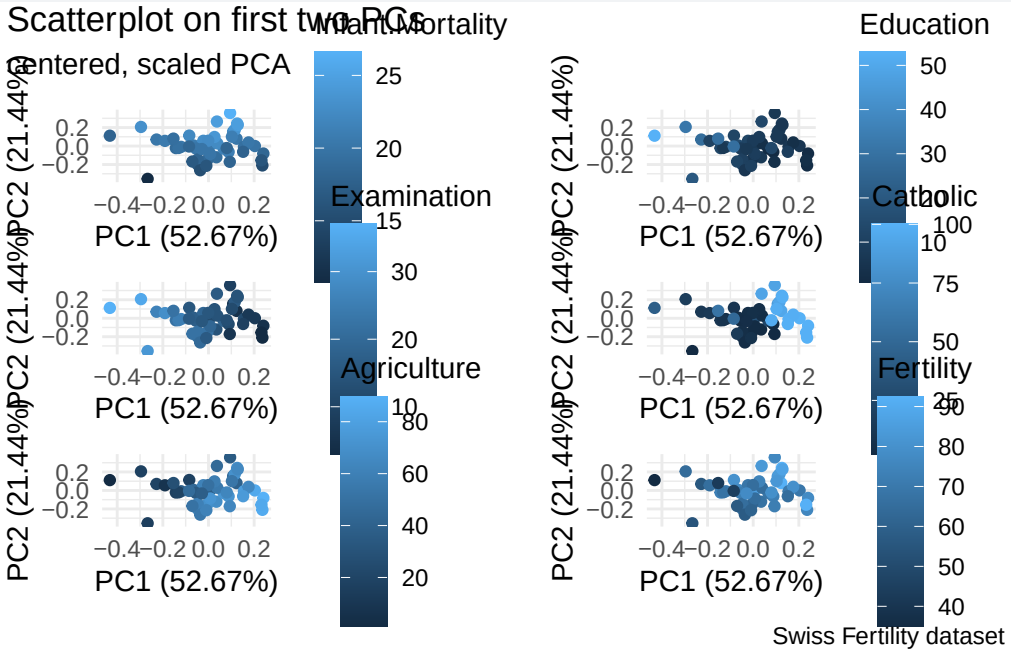
In the sequel we focus on standardized PCA.

solution

```
q <-  autoplot(pco_cs, data=swiss) +
  theme_minimal()

ts <- theme_set(theme_minimal())

(q +
  aes(color=Infant.Mortality)) +
(q +
   aes(color=Education)) +
(q +
   aes(color=Examination)) +
(q +
   aes(color=Catholic)) +
(q +
   aes(color=Agriculture)) +
(q +
   aes(color=Fertility)) +
patchwork::plot_layout(ncol = 2) +
patchwork::plot_annotation(
    title="Scatterplot on first two PCs",
    subtitle = "centered, scaled PCA",
    caption = "Swiss Fertility dataset")
```



Provide an interpretation of the first two principal axes

Question

Which variables contribute to the two first principal axes?

solution

This comes from the correlation circle. We rely on function `prep_co_circle` and on the graphical pipeline `co_circle_ppl`.

```
(
  co_circle_ppl %+%
    prep_co_circle(pco_cs) +
    ggtitle("Swiss, correlation circle",
            subtitle = "centered, scaled") +
  theme_minimal()
)
```

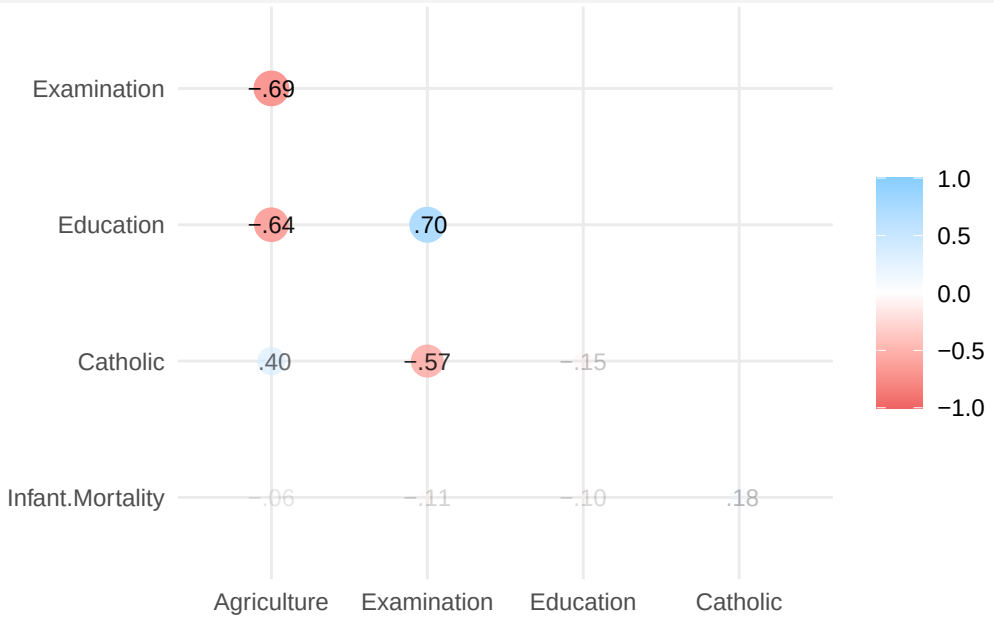
```
NULL
```

Question

Analyze the signs of correlations between variables and axes?

solution

```
swiss |>   # ggrepel::geom_text_repel(data=df_cocirc,
  #                          aes(x= 4* `1`,
  #                              y= 4 * `2`,
  #                              label=column),
  #                          color="red")
  select(-Fertility) |>
  corrr::correlate(use="pairwise.complete.obs",method="pearson", quiet=T) |>
  corrr::shave() |>
  corrr::rplot(print_cor = T) +
  theme_minimal()
```



Add the `Fertility` variable

Question

Plot again the correlation circle using the same principal axes as before, but add the `Fertility` variable.
How does `Fertility` relate with covariates? with principal axes?

solution

We use

$$D^{-1} \times U^\top \times X = V^\top$$

or equivalently

$$X^\top \times U \times D^{-1} = V$$

It is enough to multipliy the data matrix by $D^{-1} \times U^\top$ and to pipe the result into the coorelation circle graphical pipeline.

```
foo <- t(diag(svd_Y$d^(-1)) %*% t(svd_Y$u) %*% as.matrix(scale(swiss, scale=F)))

co_circle_ppl  %+% (
  as_tibble(foo) |>
  rename_with(.fn = \(x) gsub('V', '', x), .cols=everything()) |>
  mutate(row_id=rownames(foo))
) +
  theme_minimal()
```

```
NULL
```

Biplot

Question

The last svd plot (biplot) consists of overlaying the scatter plot of component `x` of the `prcomp` object and the correlation circle.
So the biplot is a graphical object built on two dataframes derived on components `x` and `rotation` of the `prcomp` objects.
Design a graphical pipeline.

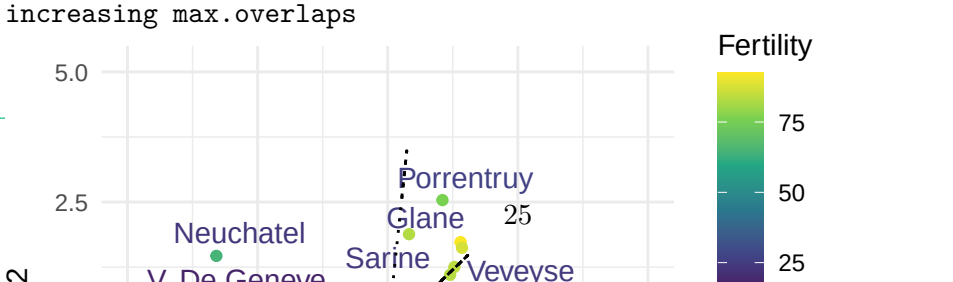
solution

```
pco <- swiss %>%
  select(-Fertility) %>%
  prcomp(scale.=T)

df_cocirc <- pco %>%
  broom::tidy(matrix="v") %>%
  tidyr::pivot_wider(id_cols =column,
              names_from = PC,
              values_from = value)

broom::augment(pco, data=swiss) %>%
  ggplot() +
  geom_point(aes(x=.fittedPC1,
                 y=.fittedPC2,
                 color=Fertility, label=.rownames)) +
  coord_fixed() +
  ggrepel::geom_text_repel(aes(x=.fittedPC1,
                               y=.fittedPC2,
                               color=Infant.Mortality,
                               label=.rownames)) +
  geom_segment(data=df_cocirc,
               mapping=aes(x= 4* `1`,
                           y= 4 * `2`,
                           linetype=factor(column),
                           label=column,
                           xend=0,
                           yend=0),
               arrow = grid::arrow(ends = "first",
                                   unit(.1, "inches")
                                  )) +
  scale_color_viridis_c() +
  xlim(c(-5,5)) +
  ylim(c(-5,5)) +
  theme_minimal()
```

```
Warning in geom_point(aes(x = .fittedPC1, y = .fittedPC2, color = Fertility, :
Ignoring unknown aesthetics: label
Warning in geom_segment(data = df_cocirc, mapping = aes(x = 4 * `1`, y = 4 * :
Ignoring unknown aesthetics: label
Warning: ggrepel: 37 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



Question

`autoplot.prcomp()` has optional arguments. If set to `True`, logical argument `loadings` overlays the scatterplot defined by the principal components with the correlation circle.

solution

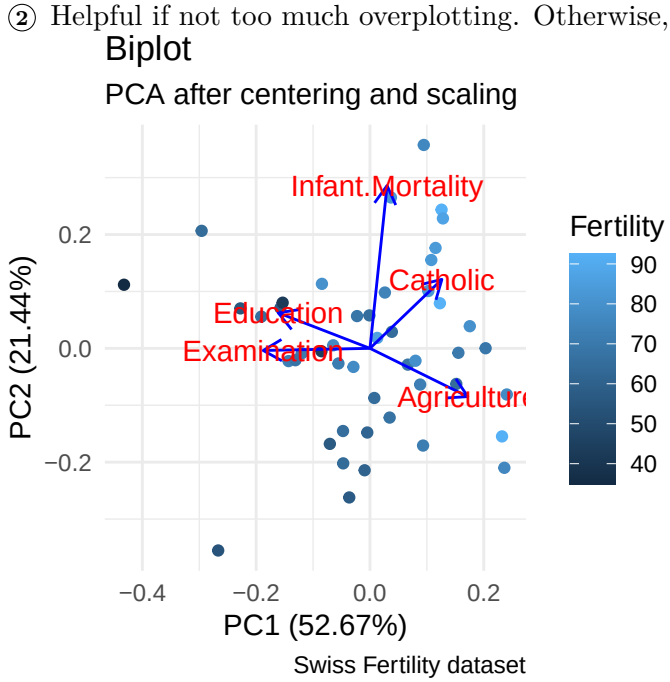
```
bip <- autoplot(pco_cs,
         data=swiss,
         color="Fertility",
         loadings = TRUE,                                    ①
         loadings.colour = 'blue',
         loadings.label = TRUE) +                            ②
  coord_fixed() +
  labs(
    title = "Biplot",
    subtitle = "PCA after centering and scaling",
    caption = "Swiss Fertility dataset"
  ) +
  theme_minimal()

ts <- theme_set(theme_minimal())

bip
```

① `loadings= TRUE` overlays the correlation circle and the principal components scatterplot.

② Helpful if not too much overplotting. Otherwise, use hovering.

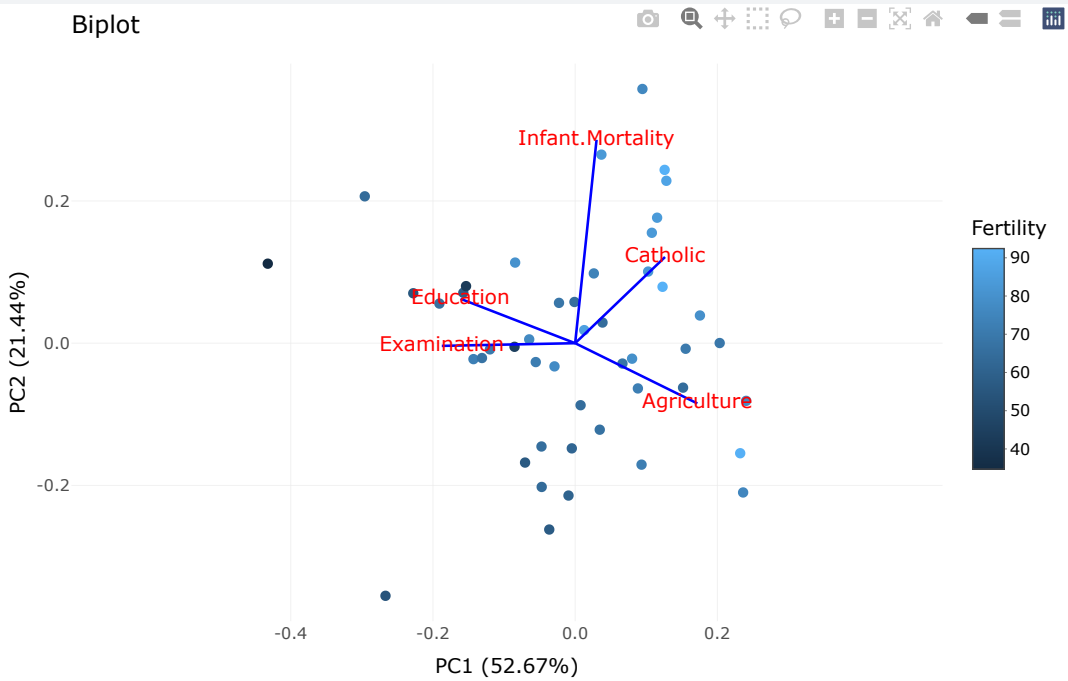


solution

Using translation to `plotly` enables hovering.

```
bip_plotly <- autoplot(pco_cs,
         data=rownames_to_column(swiss, var="district"),
         color="Fertility",
         text="district",
         loadings = TRUE,
         loadings.colour = 'blue',
         loadings.label = TRUE) +
  aes(text=district) +
  coord_fixed() +
  labs(
    title = "Biplot",
    subtitle = "PCA after centering and scaling",
    caption = "Swiss Fertility dataset"
  ) +
  theme_minimal()

bip |> plotly::ggplotly()
```



solution

Home made biplot using `broom::augment()`

```
(
broom::augment(pco, data=rownames_to_column(swiss, var="district")) |>
  ggplot() +
  geom_point(aes(x=.fittedPC1,
                 y=.fittedPC2,
                 color=Fertility)) +
  ggrepel::geom_text_repel(aes(x=.fittedPC1,
                 y=.fittedPC2,
                 label=district)) +
  coord_fixed() +
  geom_segment(data=df_cocirc,
               mapping=aes(x= 4 * `1`,
                           y= 4 * `2`,
                           linetype=factor(column),
#                           label=column,
                           xend=0,
                           yend=0),
               arrow = grid::arrow(ends = "first",
                                   unit(.1, "inches")
                                  )) +
  scale_color_viridis_c() +
  xlim(c(-5,5)) +
  ylim(c(-5,5)) +
  labs(
    title = "Biplot",
    subtitle = "PCA after centering and scaling",
    caption = "Swiss Fertility dataset"
  ) +
  theme_minimal()
) |> plotly::ggplotly()
```

Generics

`autoplot()` is an example of S3 generic function. Let us examine this function using `sloop`

Use `sloop::s3_dispatch()` to compare `autoplot(prcomp(swiss))` and `autoplot(lm(Fertility ~ ., swiss))`

solution

```r
sloop::ftype(autoplot)
```

```
[1] "S3"      "generic"
```

```r
sloop::s3_dispatch(autoplot(prcomp(swiss)))
```

```
=> autoplot.prcomp
 * autoplot.default
```

```r
sloop::s3_dispatch(autoplot(lm(Fertility ~ ., swiss)))
```

```
=> autoplot.lm
 * autoplot.default
```

Use `sloop::s3_getmethod()` to see the body of `autoplot.prcomp`

solution

```
sloop::s3_get_method(autoplot.prcomp)
## function (object, data = NULL, scale = 1, x = 1, y = 2, variance_percentage = TRUE,
##     ...)
## {
##     plot.data <- ggplot2::fortify(object, data = data)
##     plot.data$rownames <- rownames(plot.data)
##     if (is_derived_from(object, "prcomp")) {
##         ve <- object$sdev^2/sum(object$sdev^2)
##         PC <- paste0("PC", c(x, y))
##         x.column <- PC[1]
##         y.column <- PC[2]
##         loadings.column <- "rotation"
##         lam <- object$sdev[c(x, y)]
##         lam <- lam * sqrt(nrow(plot.data))
##     }
##     else if (is_derived_from(object, "princomp")) {
##         ve <- object$sdev^2/sum(object$sdev^2)
##         PC <- paste0("Comp.", c(x, y))
##         x.column <- PC[1]
##         y.column <- PC[2]
##         loadings.column <- "loadings"
##         lam <- object$sdev[c(x, y)]
##         lam <- lam * sqrt(nrow(plot.data))
##     }
##     else if (is_derived_from(object, "factanal")) {
##         if (is_null(attr(object, "covariance"))) {
```

References

S3 classes

https://scholar.google.com/citations?user=xbCKOYMAAAAJ&hl=fr&oi=ao